



PROJECT

INSTRUCTOR

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GROUP

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REPORT

1. Project Overview

This project is a full-stack web-based **Accountancy System** designed to facilitate transaction management, payments, and real-time financial reporting for organizations. Built using **HTML, CSS, JavaScript, PHP, and MySQL**, it provides secure authentication, transaction recording, payment tracking, and advanced report generation, offering both department-wise and company-wide financial visibility.

2. Functional Description

User Registration & Authentication

- Users can **register** and **log in** using a secure PHP session-based authentication system.
- Session control restricts unauthorized access to the system's internal functionalities.

Transaction Module

- Users can add **incoming** (e.g. revenue) and **outgoing** (e.g. expense) transactions.
- Transactions are linked to **departments** and **accounts**.
- Each transaction is assigned a **unique reference number** and supports descriptions and optional links to "To Accounts".

Payment Module

- Users can **record payments** against existing transactions.
- A special **Record Payment** action redirects to a filtered list of eligible transactions, making selection intuitive.
- Once a payment is made, its details are saved in the payments table, preserving metadata like amount, payment method, and timestamp.

Invoice Module

- Users can **generate invoices** from completed payments.
- Invoices are saved and accessible for later download or print.

- The system handles invoice assignment, formatting, and archiving.

Reports Module

- Offers **four key financial reports**:
 - **Profit & Loss Report** — summarizes income vs. expenses.
 - **Cash Flow Report** — tracks money in/out across time.
 - **Departmental Summary** — transaction and payment data per department.
 - **Single Department Report** — financials focused on one department.
- Reports can be filtered by **department** or cover the **whole company**.

Profile Overview

- Displays personal user information and account summary post-login.

Quick Actions Dashboard

Accessible directly after login:

- **Add Transaction**
- **Record Payment**
- **Generate Invoice**
- **Generate Reports**

Each button links to the respective module for efficient workflow.

3. Database Schema Overview

ER Diagram Explanation (Client Account System)

This ER diagram models a client account management system for an accountancy or finance platform. Here's a breakdown of the key components:

1. Entities and Attributes

Each box represents an entity (i.e., a table in a relational database), and the ovals attached to each box are attributes (i.e., columns in the table).

a. USERS

- **Attributes:**
 - UserID (Primary Key)
 - FirstName, MiddleName, LastName
 - Email, Password, PhoneNumber, Age, DateOfBirth
 - **Relationships:**
 - Has a **CLIENT ACCOUNT**
 - Belongs to a **Department**
 - Can **ask for** a **REPORT**
 - Can receive **INVOICE**
-

b. CLIENT ACCOUNT

- **Attributes:**
 - AccountID (Primary Key)
 - AccountType, Balance, Status, CreationDate
 - PaymentMethod, UserID (Foreign Key)
 - **Relationships:**
 - **Performs** TRANSACTION
-

c. TRANSACTION

- **Attributes:**
 - TransactionID (Primary Key)
 - TransactionType, TransactionDate, TransactionTime
 - Amount, PaymentMethod
 - AccountID, UserID (Both are Foreign Keys)
 - **Relationships:**
 - **Generates** REPORT
-

d. REPORTS

- **Attributes:**
 - ReportID (Primary Key)
 - DateGeneration, PeriodCovered, ReportType
-

e. DEPARTMENT

- **Attributes:**
 - DepartmentID (Primary Key)
 - DepartmentName
 - **Relationship:**
 - Linked with USERS (each user belongs to one department)
-

f. INVOICE

- **Attributes:**
 - InvoiceID (Primary Key)
 - IssueDate, DueDate, Status, TotalAmount, DiscountAmount, TaxAmount
 - **Relationships:**
 - **Received by** USERS
 - **Facilitates** PAYMENT
-

g. PAYMENT

- **Attributes:**
 - PaymentID (Primary Key)
 - PaymentDate, AmountPaid
 - **Relationships:**
 - Connected to INVOICE (pays for it)
-

Relationships and Cardinality

- A **user** has **one or more** accounts and can **perform transactions**.
 - A **transaction** is linked to both a user and a client account.
 - Each **transaction** can **generate** one or more **reports**.
 - Users can belong to a **department** and may **request reports**.
 - A **user** can receive **multiple invoices**, and each invoice can be **paid** via a payment.
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How This ER Diagram Maps to a Relational Database

In a **relational database**, each entity in the ER diagram is represented as a **table**, and attributes become **columns** in that table.

For example:

- **USERS table** will have columns like: UserID, FirstName, Email, Password, etc.
- **CLIENT_ACCOUNT table** will have: AccountID, AccountType, Balance, UserID (as a foreign key from USERS).

Foreign Keys:

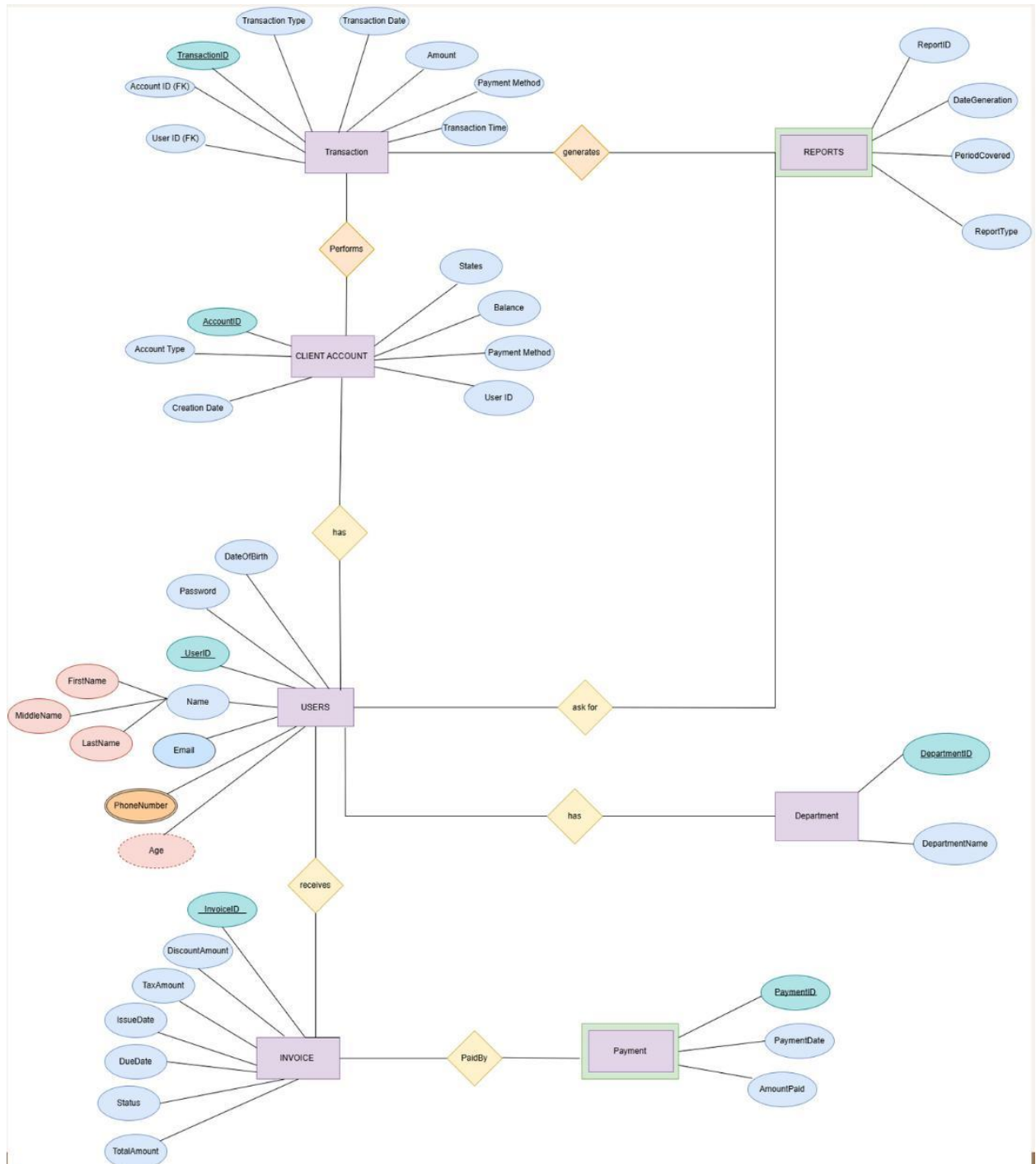
- These are used to **establish relationships** between tables. For instance:
 - The UserID in CLIENT_ACCOUNT is a foreign key referring to USERS.
 - The AccountID in TRANSACTION refers to CLIENT_ACCOUNT.

Primary Keys:

- Every table has a **Primary Key** (like UserID, AccountID, TransactionID) that **uniquely identifies** each record in that table.
-

Key Benefits of Using a Relational Database Model:

- **Data Integrity:** Maintained using foreign key constraints.
- **Scalability:** Can handle many users, accounts, and transactions efficiently.
- **Normalization:** Data is split into related tables to avoid redundancy.
- **Security:** User information and transaction data are stored separately with proper relationships.



4. Front-End Technologies

- **HTML5:** Layout structure and semantic elements.
- **CSS3:** Custom styling with responsive layouts.

- **JavaScript:** Client-side interactivity, validation, and AJAX-based content updates.
 - **Bootstrap:** Some components styled using Bootstrap classes for responsiveness.
 - **Visual Hierarchy:** Menu sidebar, dashboard cards, and modals are structured for user-friendly navigation.
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5. Backend Implementation

- **Language:** PHP (procedural + modular pattern)
 - **Database Layer:** PDO (PHP Data Objects) for secure interaction with MySQL
 - **Sessions:** Used for user authentication
 - **AJAX Handlers:** Used to fetch transactions dynamically
 - **Modular Files:**
 - config.php: DB config and credentials
 - database.php: DB connection class
 - record-transaction.php, transactions.php, record-payment.php — handle respective logic
 - reports.html and dynamic PHP endpoints for financial summaries
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6. Security Measures

- Passwords stored securely (note: hash encryption could be added)
 - Session-based login prevents unauthorized access
 - Input validation using both client-side (JavaScript) and server-side (PHP)
-

7. Testing & Validation

- **Login System** tested for invalid inputs and session hijacking
 - **Transactions** tested for:
 - Missing values
-

- Invalid amounts
 - Duplicate entries
 - **Payments** tested to ensure linkage with valid transactions
 - **Reports** validated using sample data for consistency and accuracy
-

8. Reports Sample Queries (Conceptual)

Profit/Loss Report Query:

sql

SELECT

department,

SUM(CASE WHEN transaction_type = 'incoming' THEN amount ELSE 0 END) AS
income,

SUM(CASE WHEN transaction_type = 'outgoing' THEN amount ELSE 0 END) AS
expense

FROM transactions

GROUP BY department;

Cash Flow:

SELECT

DATE(created_at) as date,

SUM(CASE WHEN transaction_type = 'incoming' THEN amount ELSE 0 END) AS
inflow,

SUM(CASE WHEN transaction_type = 'outgoing' THEN amount ELSE 0 END) AS
outflow

FROM transactions

GROUP BY DATE(created_at);

9. Challenges & Resolutions

Challenge

Resolution

Associating payments with transactions Used transaction IDs and reference linking

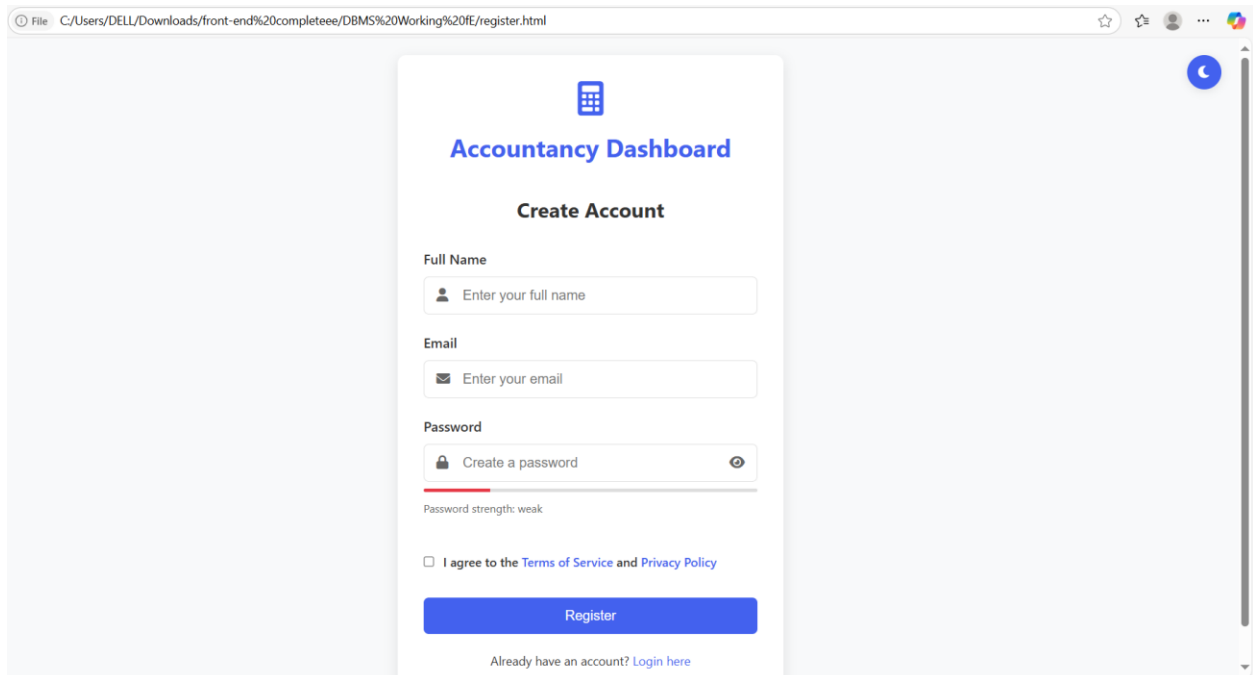
Report filtering and summarization Used conditional SQL aggregation

Real-time invoice creation Payment → invoice mapping using PHP

Keeping UI consistent Created reusable components in /components/

SNIPPETS

Login Page:



The screenshot shows a web browser window displaying the 'Accountancy Dashboard' 'Create Account' page. The page has a light blue header with a calculator icon and the title 'Accountancy Dashboard'. Below the title is the 'Create Account' section. It contains three input fields: 'Full Name' with a person icon, 'Email' with an envelope icon, and 'Password' with a lock icon. The password field has a strength indicator showing 'Password strength: weak'. Below the fields is a checkbox for 'I agree to the Terms of Service and Privacy Policy'. At the bottom of the form is a blue 'Register' button. A link 'Already have an account? Login here' is at the very bottom.

File C:/Users/DELL/Downloads/front-end%20completeee/DBMS%20Working%20FE/register.html

Accountancy Dashboard

Create Account

Full Name
Enter your full name

Email
Enter your email

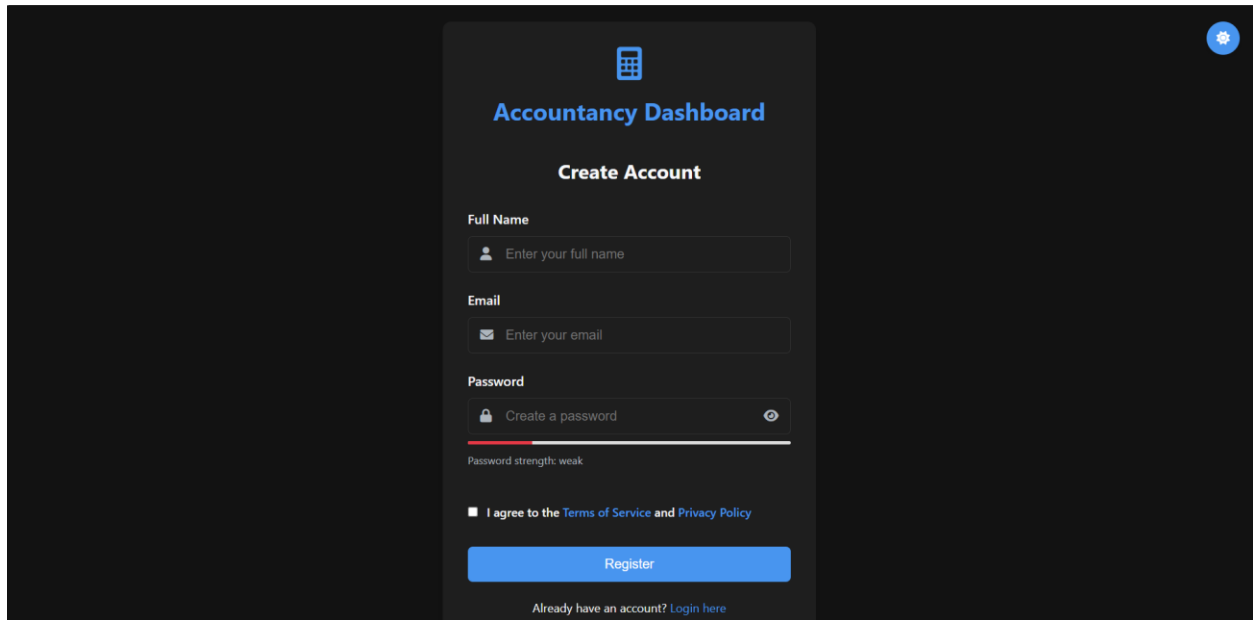
Password
Create a password

Password strength: weak

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Register

Already have an account? [Login here](#)



The image shows a dark-themed web interface for an "Accountancy Dashboard". At the top center is a blue icon of a calculator. Below it, the text "Accountancy Dashboard" is displayed in blue. Underneath, the heading "Create Account" is centered. The form contains three input fields: "Full Name" with a person icon, "Email" with an envelope icon, and "Password" with a lock icon. The password field has a strength indicator bar that is mostly red, labeled "Password strength: weak". Below the password field is a checkbox labeled "I agree to the Terms of Service and Privacy Policy". At the bottom of the form is a blue "Register" button. Below the button, there is a link: "Already have an account? Login here". A blue gear icon is located in the top right corner of the dashboard area.

Accountancy Dashboard

Create Account

Full Name
Enter your full name

Email
Enter your email

Password
Create a password

Password strength: weak

☐ I agree to the Terms of Service and Privacy Policy

Register

[Already have an account? Login here](#)

Crediantials Entry:



Accountancy Dashboard

Create Account

Full Name

 test

Email

 tes@gmail.com

Password

 123456



Password strength: weak

☒ I agree to the [Terms of Service](#) and [Privacy Policy](#)

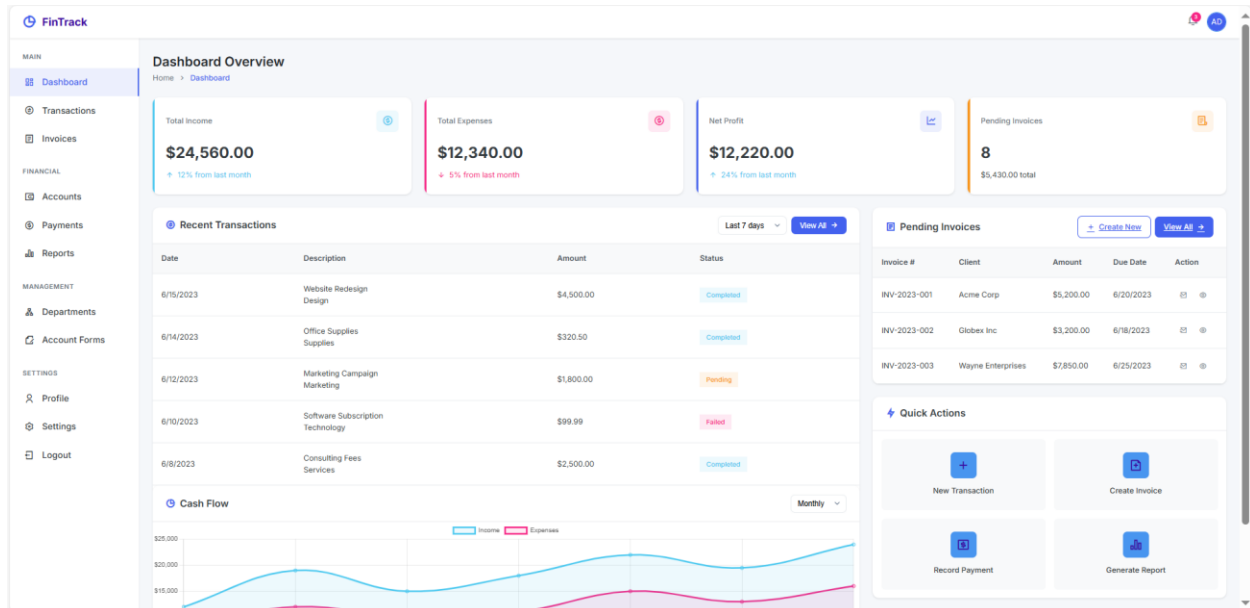
Register

Already have an account? [Login here](#)

Database Updation:

													id	email	password	first_name	last_name	phone	role	status	created_at	update
☐	 Edit	 Copy	 Delete	1	admin@accountancy.com	\$2y\$10\$K4oqYoRRFqEDn1euHe446eTsignPz5ZGYeTQfHhSkmQ...	System	Administrator	NULL	admin	active	2025-06-15 06:32:31	2025-06-15 06:32:31									
☐	 Edit	 Copy	 Delete	2	new@gmail.com	\$2y\$10\$K4oqYoRRFqEDn1euHe446eTsignPz5ZGYeTQfHhSkmQ...	new	NULL	NULL	user	active	2025-06-19 17:05:12	2025-06-19 17:05:12									

Dashboard:



New Transaction:

Record New Transaction

Transaction Type

Outgoing



Department

IT



Transaction Name

New

Amount

100

Due Date

06/21/2025



Description

New



Vendor/Supplier

Aliyan

Record Transaction

Transactions list:

All Transactions							
Type:	All Types	Department:	All Departments	Status:	All Statuses		
ID	Type	Name	Department	Amount	Due Date	Status	Actions
TXN-1750021476438	Outgoing	Assistance	IT	\$6000.00	Jun 18, 2025	Paid	View Delete
TXN-1750367906087	Outgoing	New	IT	\$100.00	Jun 20, 2025	Pending	Pay View Delete

Delete Operation:

All Transactions							
Type:	All Types	Department:	All Departments	Status:	All Statuses		
ID	Type	Name	Department	Amount	Due Date	Status	Actions
TXN-1750367906087	Outgoing	New	IT	\$100.00	Jun 20, 2025	Pending	Pay View Delete

Can view:

Type: All Types

Department: All Departments

ID	Type	Name
TXN-1750367906087	Outgoing	New

Pay

View

Delete

This page says

Transaction Details:

ID: TXN-1750367906087

Type: Outgoing

Name: New

Department: IT

Amount: \$100.00

Due Date: Jun 20, 2025

Status: Pending

Vendor: Aliyan

Description: New

OK

Payment:

By clicking pay, entered to this page for payment:

New

Department: IT

Vendor: Aliyan

This page says

Payment recorded successfully!

OK

h 20, 2025

Payment Amount

100.00

Payment Date

06/19/2025

Payment Method

Credit Card

Reference Number

1

Notes

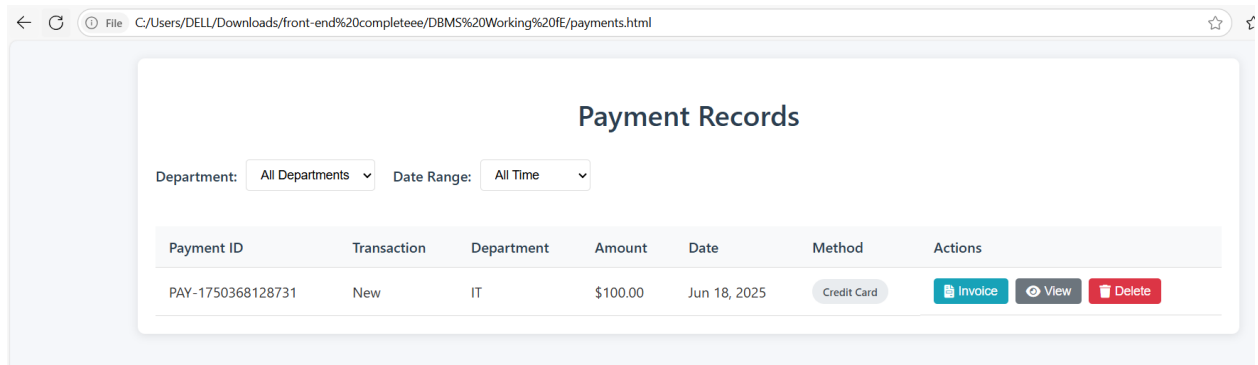
New

Record Payment

Database Updation:

	id	account_id	transaction_type	amount	description	reference_number	to_account_id	status	created_at	updated_at
☐	1	1	credit	100.00	New	1	1	completed	2025-06-15 15:55:03	2025-06-19 17:20:09

List of payments:



Database Updation:

	id	invoice_id	transaction_id	amount	payment_method	payment_date	reference_number	notes	status	created_at	updated_at
☐ Edit ☐ Copy ☐ Delete	1	1	1	100.00	card	0000-00-00	1	New	completed	2025-06-19 17:33:11	2025-06-19 17:33:11

View Invoice:

Generate Invoice

INVOICE

IT Department

Invoice #: INV-1750368631017

Date: Jun 18, 2025

Payment ID: PAY-1750368128731

Transaction ID: TXN-1750367906087

From:

Company Name

123 Business Street

City, State 10001

Phone: (123) 456-7890

Email: accounts@company.com

To:

Aliyan

Vendor Address

Vendor City, State

Reference: 1

Description	Amount
Payment for New	\$100.00
Subtotal:	\$100.00
Tax (0%):	\$0.00
Total:	\$100.00

Payment Method: Credit Card | Paid on Jun 18, 2025

Thank you for your business!

Download PDF

Print Invoice

Save Invoice

List of invoices:

Saved Invoices

Department: All Departments ▼ Date Range: All Time ▼

Invoice #	Payment ID	Department	Amount	Date	Actions
INV-1750368631017	PAY-1750368128731	IT	\$100.00	Jun 18, 2025	View PDF Delete

Report Generation:

Financial Reports

Report Type: Profit & Loss ▼ Time Period: All Time ▼ Department: All Departments ▼ [Generate Report](#) [Export](#)

Select report criteria and click "Generate Report"

Financial Reports

Report Type:

Profit & Loss

Time Period:

All Time

Department:

All Departments

Generate Report

Export

Total Income

\$0.00

From 0 transactions

Total Expenses

\$100.00

From 1 transactions

Net Profit

\$100.00

Loss

Organization Profit & Loss Report

Category	Amount	% of Total
Total Income	\$0.00	100%
Total Expenses	\$100.00	N/A
Net Profit/Loss	\$100.00	N/A

Departmental Breakdown

Department	Income	Expenses	Net
HR	\$0.00	\$0.00	\$0.00
IT	\$0.00	\$100.00	\$100.00
MARKETING	\$0.00	\$0.00	\$0.00
FINANCE	\$0.00	\$0.00	\$0.00

Profit & Loss

Profit & Loss



12. Conclusion

The Web-Based Accountancy System fulfills all requirements of the open-ended lab with a complete feature set, logical flow, and dynamic reporting engine. With clean design, secure backend, and real-world extensibility (like PDF exports, graphs), this system demonstrates not only technical competence but also a practical understanding of database-driven applications.